Zhejiang University-University of Illinois Urbana-Champaign Institute

(ZJU-UIUC INSTITUTE)

Electrical Engineering and Automation Curriculum

(Updated in August 2025)

1. Program Educational Objectives

Electrical engineering and Automation is a multifaceted discipline that over the last century has produced an astounding progression of technological innovations that have shaped virtually every aspect of modern life. Electrical engineers need a broad and solid foundation in mathematics and physics to support their education in the engineering principles of analysis, synthesis, design, implementation, and testing of the devices and systems that provide the bedrock of modern energy, communication, sensing, computing, medical, security, and defense infrastructures. Within each sub-discipline one can find application domains that strongly rely on hands-on experimental work or that are based on theoretical, mathematical, and computational approaches. The multidisciplinary nature of electrical engineering education addresses the growing demand for the innovation and design of sensing, communication, computing, and decision-making systems of increasing complexity in consumer, defense, and medical applications.

2. Graduation Requirement

The curriculum leading to the degree of Bachelor of Engineering from ZJUI requires 137 credits and is organized into required courses; technical electives; liberal education; and other electives.

3. Required Courses

3.1 Orientation and Professional Development

These courses introduce the opportunities and resources our institute and curriculum can offer you as you work to achieve your career goals. They also provide the skills to work effectively and successfully in the engineering profession.

Course Code	Course Name	Credit
ENG 100	Engineering Orientation	1

3.2 Foundational Mathematics and Science

These courses stress the basic mathematical and scientific principles upon which the engineering discipline is based.

Course Code	Course Name	2020	2021-
MATH 221	Calculus Ⅰ	4	4
MATH 231	Calculus Ⅱ	3	3
MATH 241	Calculus Ⅲ	4	4
MATH 257	Linear Algebra with Computational Application		3
MATH 285	Introduction to Differential Equation		3
MATH 286	Intro to Differential Equations Plus	4	
CHEM 102	General Chemistry Ⅰ	3	3
CHEM 103	General Chemistry Lab Ⅰ	1	1
PHYS 211	University Physics: Mechanics	4	4
PHYS 212	University Physics: Elec & Mag	4	4
PHYS 213	University Physics: Thermal Physics	2	2
PHYS 214	University Physics: Quantum Physics	2	2
	Total required	31	33

3.3 Technical Core

These courses stress fundamental concepts and basic laboratory techniques that comprise the common intellectual understanding of electrical engineering.

Course Code	Course Name	2020-
ECE 110	Intro to Electronics	3
ECE 120	Intro to Computing	4
ECE 220	Computer Systems & Programming	4
ECE 210	Analog Signal Processing	4
ECE 329	Fields and Waves Ⅰ	3
ECE 313	Probability with Engineering Application	3
ECE 385	Digital Systems Laboratory	3
ECE 340	Semiconductor Electronics	3
ECE 445	Senior Design Project Lab	4
	Total required	31

4 Elective Courses

4.1 Technical Electives

This elective requirement gives each student the freedom to develop a technical course of study in electrical engineering of considerable breadth and focus. The Advanced Core ECE Electives are introductory to major sub-disciplines of electrical engineering, such as electrical and computer engineering: bioengineering, acoustics, and magnetic resonance engineering; circuits and signal processing; communication and control; computer engineering; electromagnetics, optics, and remote sensing; microelectronics and quantum electronics; power and energy systems.

Students must complete a minimum of total required hours on technical electives, which include at least:

1) 2 courses Non-ECE courses, see below section 4.1.1 for detail.

2)	3 courses	Selected from the following list of **Advanced Core ECE** electives, see below section 4.1.2 for detail.	
3)	3 courses	ECE Labs, see below section 4.1.3 for detail.	
4)		ECE courses, see below section 4.1.4 for detail.	
		Total required	30

4.1.1 Non-ECE Courses

Students are encouraged to build up their cross-discipline study by taking non-ECE technical courses from Civil Engineering, Mechanical Engineering, Computer Science, and others at ZJUI. Two courses are needed.

Course Code	Course Name	Credits
CEE 310	Transportation Engineering	3
ME 200	Thermodynamics	3
ME 320	Heat Transfer	4
ME 371	Mechanical Design II	3
ME 400	Energy Conversion Systems	3 or 4
ME 472	Introduction to Tribology	3
ME 412	Numerical Thermo-Fluid Mechs	2 to 4
ME 485	MEMS Devices & Systems	3
ME 498VS	Vacuum Engineering in Semiconductor Industry	3
TAM 211	Statics	3
TAM 212	Introductory Dynamics	3
TAM 251	Introductory Solid Mechanics	3
TAM 335	Introductory Fluid Mechanics	4
CS 101	Introduction to Computing: Engineering & Science	3
CS 225	Data Structures	4
CS 357	Numerical Methods I	3
CS 412	Introduction to Data Mining	3 or 4
CS 442	Trustworthy machine learning	3

CS 450	Numerical Analysis	3 or 4
ABE 436	Renewable Energy Systems	3
CHEM 104	General Chemistry II	3
CHEM 105	General Chemistry Lab II	1
MATH 213	Introduction to Discrete Mathematics	3
STAT 432	Basics of Statistical Learning	3

4.1.2 Advanced Core ECE

Select three courses from the list below:

Course Code	Course Name	Credits
ECE 391	Computer Systems Engineering **or** CS 225 - Data Structure	4
ECE 310	Digital Signal Processing	3
ECE 330	Power Circuits & Electromechanics	3
ECE 342	Electronic Circuits	3
ECE 350	Fields and Waves II	3

4.1.3 ECE Labs

Students must take 3 labs, at least one of which must be a hardware lab.

1) Hardware Labs

Course Code	Course Name	Credits
ECE 343	Electronic Circuits Laboratory	1
ECE 438	Communication Networks	3 or 4
ECE 469	Power Electronics Laboratory	2
ECE 470	Introduction to Robotics	4
ECE 486	Control Systems	4

2) Software Labs

Course Code	Course Name	Credits
ECE 311	Digital Signal Processing Lab	1
ECE 314	Probability in Engineering Lab	1
ECE 365	Data Science and Engineering	3

ECE 391	Computer Systems Engineering	4
ECE 364	Programing Methods for Machine Learning	3

4.1.4 ECE courses

Course Code	Course Name	Credits
ECE 310	Digital Signal Processing	3
ECE 311	Digital Signal Processing Lab	1
ECE 314	Probability in Engineering Lab	1
ECE 342	Electronic Circuits	3
ECE 343	Electronic Circuits Laboratory	1
ECE 350	Fields and Waves II	3
ECE 365	Data Science and Engineering	3
ECE 374	Introduction to Algorithms & Models of Computation	4
ECE 391	Computer Systems Engineering	4
ECE 416	Biosensors	3
ECE 441	Physics & Modeling Semiconductor Device	3
ECE 448	Artificial Intelligence	3 or 4
ECE 449	Machine Learning	3
ECE 453	Wireless Communication Systems	4
ECE 457	Microwave Devices & Circuits	3
ECE 459	Communications Systems	3
ECE 464	Power Electronics	3
ECE 469	Power Electronics Laboratory	2
ECE 470	Introduction to Robotics	4
ECE 471	Data Science Analytics using Probabilistic Graph Models	3
ECE 476	Power System Analysis	3
ECE 485	MEMS Devices & Systems	3
ECE 486	Control Systems	4
ECE 491	Numerical Analysis	3
ECE 498MF	Mathematical Foundations for Machine Learning	3
ECE 498EAI	Embodied Artificial Intelligence	3

4.2 Liberal Education

The liberal education courses develop students' understanding of human culture and society, build skills of inquiry and critical thinking, and lay a foundation for civic engagement and lifelong learning. To satisfy the General Education requirements, students must take all courses for grade and complete courses based on the table below:

Category	Credit
1) Humanities & Arts (2 courses)	6
2) Social & Behavioral Sciences (2 courses)	6
3) Culture Studies	6
Western/Comparative Culture(s) (1 course) or	
Non-Western Culture(s) (1 course)	
Total required	**18**

To meet the General Education requirements, students need to select courses that fulfill these four subcategories and accumulate a total of 18 credits. Properly chosen courses in the Social and Behavioral Sciences and Humanities & Arts categories can also satisfy the Western and non-Western culture requirements, ensuring that students receive a comprehensive education.

4.3 Free Electives

These unrestricted electives give the student the opportunity to explore any intellectual area of unique interest. This freedom plays a critical role in helping students to define research specialties. Students are encouraged to take cross-discipline courses as free electives.

Free electives	2020	2021-2022	2023-
Total	4	2	1

5 ZJU Required Liberal Education

These courses introduce Chinese modern history, social development, government policies, etc., help students to improve their English and maintain a healthy lifestyle.
International students need to complete Chinese language study and Chinese society to fulfill the graduate requirement.

Course Code	Course Name	2020 Credits	2021-2022 Credits	2023- Credits
CHIN 1001	Chinese I	4	4	4
CHIN 1002	Chinese II	5	5	4
CHIN 1003	Chinese III	4	4	4
CHIN 1004	Chinese IV	4	4	4
CHIN1005	International Chinese Language Proficiency and Testing			2
CULT 2001	China Survey	3	3	3
ST 4001	Senior Thesis		2	2
	Total	20	22	23